High Torque Timing Pulleys - S5M

Compatible with S2M type from Mitsuboshi Belting Ltd. as well as Bando Chemical Industries Ltd.
Compatible with S5M type from Mitsuboshi Belting Ltd. as well as Bando Chemical Industries Ltd.

■For High Torque Timing Belts, see P.1465 and for Idlers with Teeth, see P.1433. For High Torque Timing Belts, see P.1451.



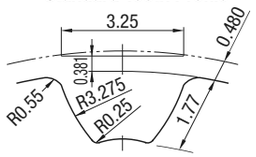
RoHS

Type	Belt Width 10mm S5M100	15mm S5M150	25mm S5M250	M Material *1 Pulley	Flange	S Surface Treatment	A Accessory *1 Set Screws
HTPA	●	●	●	A2000 Series Aluminum Alloy	Aluminum Alloy	Clear Anodize	EN 1.4301 Equiv.
HTPB	●	●	●			Black Anodize	
HTPK	●	●	●			Hard Clear Anodize *2	
HTPN	●	●	●			Electroless Nickel Plating	
HTPT	●	●	●	EN 1.1191 Equiv.	EN 1.0330 Equiv.	-	EN 1.7220 Equiv. (Black Oxide)
HTPM	●	●	●			Black Oxide	
HTPP	●	●	●			Electroless Nickel Plating	
HTPS	●	●	-	EN 1.4301 Equiv.	EN 1.4301 Equiv.	-	EN 1.4301 Equiv.

Flange is installed, and set screws are included with Shaft Bores P, N and C.

*1. The above material and accessory might be changed to the ones equivalent to the originals.
*2. Hard Clear Anodize: Film Hardness 300HV ~

Standard Tooth Profile

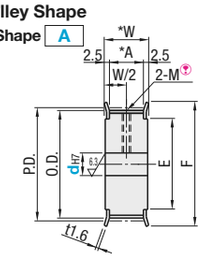


Tooth groove dimensions slightly vary according to the number of teeth.

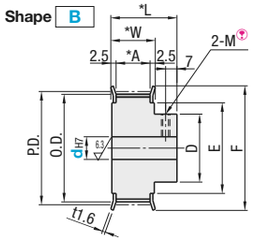
(Pitch:5.0mm)

• Pulley Shape

Shape A



Shape B



■Tapped Hole Dimensions (Shaft Bore Specs.: P, N, C)

dH7 Shaft Bore I.D.	M (Coarse)	Accessory: Set Screw
5	M3	M3x3
6~12	M4	M4x3
13~17	M5	M5x4
18~30	M6	M6x5
31~45	M8	M8x6
46~65	M10	M10x8

Shaft Bore Specs. H (Round hole), V or F (Stepped Hole) and Y (Both Sides Stepped Hole) do not have tapped holes.

■Number of Teeth / Dimension

mm	Number of Teeth 14	15	16	18	19	20	22	24	25	26	28	30	32	34	36	40	44	48	50	60	72
P.D.	22.28	23.87	25.46	28.65	30.24	31.83	35.01	38.20	39.79	41.38	44.56	47.75	50.93	54.11	57.30	63.66	70.03	76.39	79.58	95.49	114.59
O.D.	21.32	22.91	24.50	27.69	29.28	30.87	34.05	37.24	38.83	40.42	43.60	46.79	49.97	53.15	56.34	62.70	69.07	75.43	78.62	94.53	113.63
D	14	15	17	19	19	19	24	27	27	31	32	33	37	40	40	47	50	60	63	75	90
F	26	28	32	33	36	36	40	45	45	48	48	52	55	61	61	67	74	83	87	99	119
E	16	18	20	22	24	24	27	30	30	35	35	36	40	45	45	50	58	63	67	80	100

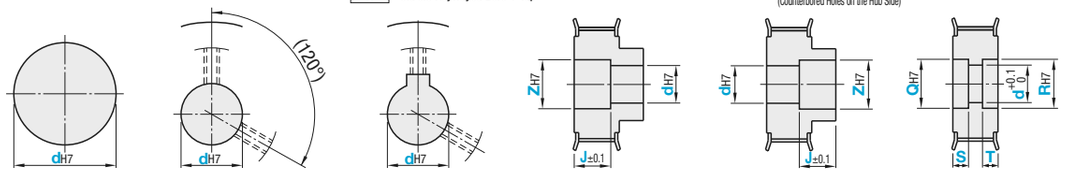
■Belt Nominal Width / Dimension

mm	Nominal S5M100	S5M150	S5M250
A	11	17	27
W	16	22	32
L	28	34	44

• Shaft Bore Specs.

Surface treatment may not be applied to shaft bores.

- H Round Hole
- P Round Hole + Tap
- N New JIS Keywayed Bore + Tap
- C Old JIS Keywayed Bore + Tap
- V Stepped Hole
- F Stepped Hole (Counterbored Holes on the Hub Side)
- Y Both Ends Stepped Hole



- No tapped holes or set screws.
- For A-Shape pulley, the screw holes are set at around 120° to keep away from peaks.
- For Keyway Dimension Details, see P.1377. When selecting the shaft bore dia. 10 and the keyway width 4.0mm (height 1.8mm) for New JIS Keywayed Bore, specify NK10.
- No tapped holes or set screws.
- Applicable to Shape B only.
- No tapped holes or set screws.
- Applicable to Shape A only.
- Shaft Bore Dia. d is +0.1 / 0
- No tapped holes or set screws.

Part Number Type	Number of Teeth	Type Nominal Width	Pulley Shape
Aluminum HTPA HTPB HTPK HTPN	14	S5M100	A B
	15		
	16		
	18		
	19		
	20		
	22		
	24		
	25		
Steel HTPT HTPM HTPP	26	S5M150	
	28		
	30	S5M250	
	32		
Stainless Steel HTPS	34		
	36		
	40		
	44		
	48		
	50		
	60		
	72		

Pulley Shape A

Shaft Bore Specifications (–): Specify in 1mm Increment, (,): Select the former or latter

H Round Hole	P Round Hole + Tap	N, C Keyway + Tap	V Stepped Hole V	V Stepped Hole Z d±0.2	V Stepped Hole J (0.1mm Increment)	Y Both Ends Stepped Y	Y Both Ends Stepped Q, R Q(R)-d±0.2	Y Both Ends Stepped S, T
5, 6, 6.35, 7~10	5, 6, 6.35, 7~10	8, 10, NK10	5, 6	7, 8	2.0≤J≤W-2.0	5, 6	7, 8	3~14 S+T≤W-3
5, 6, 6.35, 7~10	5, 6, 6.35, 7~10	8, 10, NK10	5~8	7~10		5~8	7~10	
5, 6, 6.35, 7~12	5, 6, 6.35, 7~12	8, 10, NK10, 11, 12	5~10	7~12		5~10	7~12	
6, 6.35, 7~14	6, 6.35, 7~12	8, 10, NK10, 11, 12	6~12	8~14		6~12	8~14	
6, 6.35, 7~16	6, 6.35, 7~16	8, 10, NK10, 11~16	6~14	8~16		6~14	8~16	
6, 6.35, 7~16	6, 6.35, 7~16	8, 10, NK10, 11~16	6~14	8~16		6~14	8~16	
7~19	7~18	8, 10, NK10, 11~18	7~17	9~19		7~17	9~19	
7~22	7~20	8, 10, NK10, 11~20	7~20	9~23		7~20	9~22	
7~22	7~20	8, 10, NK10, 11~20	7~20	9~23		7~20	9~22	
8~27	8~22	8, 10, NK10, 11~22	8~25	10~27		8~25	10~27	
8~27	8~24	8, 10, NK10, 11~24	8~25	10~27		8~25	10~27	
10~28	10~26	10, NK10, 11~26	10~26	12~28		10~26	12~28	
10~32	10~28	10, NK10, 11~28	10~30	12~32		10~30	12~32	
10~37	10~30	10, NK10, 11~30	10~35	12~37		10~35	12~37	
10~37	10~30	10, NK10, 11~30	10~35	12~37		10~35	12~37	
10~42	10~38	10, NK10, 11~38	10~40	12~42		10~40	12~42	
12~50	12~42	12~40	12~48	14~50		12~48	14~50	
12~55	12~45	12~40	12~53	14~55		12~53	14~55	
12~59	12~45	12~43	12~57	14~59		12~57	14~59	
12~72	12~45	12~45	12~70	14~72		12~70	14~72	
12~80	12~65	12~50	12~80	14~92		12~75	14~92	

Pulley Shape B

Shaft Bore Specifications (–): Specify in 1mm Increment, (,): Select the former or latter

H Round Hole	P Round Hole + Tap	N, C Keyway + Tap	V, F Stepped Hole V, F	V, F Stepped Hole Z d±0.2	V, F Stepped Hole J (0.1mm Increment)
5, 6, 6.35, 7~10	5, 6, 6.35, 7, 8	-	5, 6	7, 8	2.0≤J≤L-2.0
5, 6, 6.35, 7~10	5, 6, 6.35, 7, 8		5~8	7~10	
5, 6, 6.35, 7~12	5, 6, 6.35, 7~10	8	5~10	7~12	
6, 6.35, 7~14	6, 6.35, 7~11	8, 10, NK10	6~12	8~14	
6, 6.35, 7~15	6, 6.35, 7~11	8, 10, NK10	6~13	8~15	
6, 6.35, 7~15	6, 6.35, 7~11	8, 10, NK10	6~13	8~15	
7~19	7~15	8, 10, NK10, 11, 12	7~17	9~19	
7~22	7~17	8, 10, NK10, 11~13	7~20	9~23	
7~22	7~17	8, 10, NK10, 11~15	7~20	9~23	
8~27	8~21	8, 10, NK10, 11~17	8~25	10~27	
8~27	8~22	8, 10, NK10, 11~18	8~25	10~27	
10~28	10~23	10, NK10, 11~18	10~26	12~28	
10~32	10~27	10, NK10, 11~22	10~30	12~32	
10~36	10~30	10, NK10, 11~25	10~34	12~36	
10~36	10~30	10, NK10, 11~25	10~34	12~36	
10~42	10~35	10, NK10, 11~28	10~40	12~42	
12~46	12~38	12~32	12~44	14~46	
12~55	12~45	12~40	12~53	14~55	
12~59	12~45	12~43	12~57	14~59	
12~71	12~45	12~45	12~69	14~71	
12~80	12~65	12~50	12~80	14~86	

Shaft Bore Dia. 8, 11, 13, 14, 17, 21~50 are not available for Shaft Bore Spec. C.

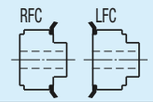
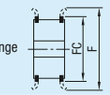
Ordering Example

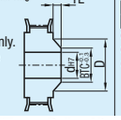
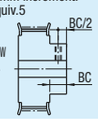
	Part Number	Pulley Shape	Shaft Bore Specs., I.D.	Z	J	Q	R	S	T
(Shaft Bore Specs.: H, P, N, C)	HTPA20S5M150	B	NK10						
(Shaft Bore Specs.: V, F)	HTPA26S5M150	A	V10	Z23	J16.0				
(Shaft Bore Specs: Y)	HTPA40S5M250	A	Y17			Q35	R35	S10	T10

Number of Teeth	Body Price HTPA (x1.0) S5M100 Shape A	HTPA (x1.0) S5M100 Shape B	HTPB, HTPK (x1.1) S5M150 Shape A	HTPB, HTPK (x1.1) S5M150 Shape B	HTPN (x1.2) S5M250 Shape A	HTPN (x1.2) S5M250 Shape B	HTPT (x1.0) S5M100 Shape A	HTPT (x1.0) S5M100 Shape B	HTPM (x1.05) S5M150 Shape A	HTPM (x1.05) S5M150 Shape B	HTPP (x1.15) S5M250 Shape A	HTPP (x1.15) S5M250 Shape B	HTPS (x1.0) S5M100 Shape A	HTPS (x1.0) S5M100 Shape B	HTPS (x1.0) S5M150 Shape A	HTPS (x1.0) S5M150 Shape B	Shaft Bore Machining Charge (Body Price +) HTPA, HTPB, HTPK, HTPN HTPT, HTPM, HTPP P Hole	N, C, V, F Hole	Y Hole	HTPS P Hole	N, C, V, F Hole	Y Hole
14																						
15																						
16																						
18																						
19																						
20																						
22																						
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44																						
48																						
50																						
60																						
72																						

Alterations

	Part Number	Pulley Shape	Shaft Bore Specs., I.D.	Z	J	Q	R	S	T	(KC90···etc.)
	HTPM60S5M150	A	H30							KSC50

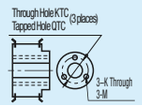
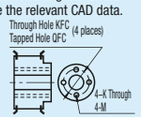
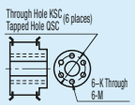
Alterations	Set Screw Angle	No Flange	Single Flange	Flange Cut
Code	KC90	NFC	RFC, LFC	FC
Spec.	Changes an angle of set screw to 90. For A-Shape pulley, the screw holes are set at around 90° to keep away from peaks. 	(Flange 2 pcs. Included) Ordering Code NFC 	(Flange 1 pc. Included) Ordering Code RFC 	Cut the flange O.D. in 0.5mm increment. Ordering Code FC17 Application Notes: FC≥(O.D.)+1; FC≤F-2; No surface treatment is applied on flange circumference; ⊗Not available for Stainless Steel Type. 

Alterations	Adds taper for retaining bearing	Hub Shortening	Tapped Hole Dimensions	Changes the length of the included set screws
Code	BTC	BC	TPC	SLH
Spec.	Add taper for retaining bearing inner ring Ordering Code BTC6-TL1.5 Application Notes: Applicable to Shape A only; Applicable to Shape Bore Specs. H and P only; TL<L-W 	Cuts the hub length in 0.5mm increment. Ordering Code EN CC491K Equiv.5 Application Notes: Shaft Bore Specs. H, V, F: 3≤BC≤L-W; Shaft Bore Specs. P, N, C: M+3≤BC≤L-W; ⊗Not available for Shape A. 	Ordering Code TPC5 Application Notes: Applicable to Shaft Bore Specs. P, N, C only.	Ordering Code SLH10 Application Notes: Applicable to Shaft Bore Specs. P, N, C only.

M	TPC
M3	M4
M4	M3, M5
M5	M4, M6
M6	M5, M8
M8	M6

Set Screws	SLH
M3x3	6
M4x3	5, 8
M5x4	6, 10
M6x5	10
M8x6	10, 12
M10x8	12, 15

For details, see the "Timing Pulley Alterations - Overview" section (P.1378).

Alterations	Side Through Hole / Side Tapped Hole, 3 places	Side Through Hole / Side Tapped Hole, 4 places	Side Through Hole / Side Tapped Hole, 6 places
Code	KTC, QTC	KFC, QFC	KSC, QSC
Spec.	Machine Through Hole / Tapped Hole on the side surface of hub side. Ordering Code (Through Hole) KTC20-K5.0; Ordering Code (Tapped Hole) QTC28-M4; Selection (Through Hole) K Selection K4.0~K13.0 (0.5mm Increment); Selection (Tapped Hole) M Selection M3, M4, M5, M6, M8. Application Notes: ⊗Not applicable to Shaft Bore Specs. F or Y; ⊗When KTC/QTC is selected for Shaft Bore Specs. P, N and C, KC90 is not available. 	Machine Through Hole / Tapped Hole on the side surface of hub side. Ordering Code (Through Hole) KFC20-K5.0; Ordering Code (Tapped Hole) QFC28-M4; Selection (Through Hole) K Selection K4.0~K13.0 (0.5mm Increment); Selection (Tapped Hole) M Selection M3, M4, M5, M6, M8. Application Notes: ⊗Not applicable to Shaft Bore Specs. F or Y; Specify KC90 when selecting KFC/QFC for Shaft Bore Specs. P, N and C; Side holes and tooth side tapped holes might interfere with each other. For details, see the relevant CAD data. 	Machine Through Hole / Tapped Hole on the side surface of hub side. Ordering Code (Through Hole) KSC20-K5.0; Ordering Code (Tapped Hole) QSC28-M4; Selection (Through Hole) K Selection K4.0~K13.0 (0.5mm Increment); Selection (Tapped Hole) M Selection M3, M4, M5, M6, M8. Application Notes: ⊗Not applicable to Shaft Bore Specs. F or Y; ⊗KSC/QSC is not applicable to the Shaft Bore Specs. P, N and C. 

For details, see the "Timing Pulley Alterations - Overview" section (P.1378).